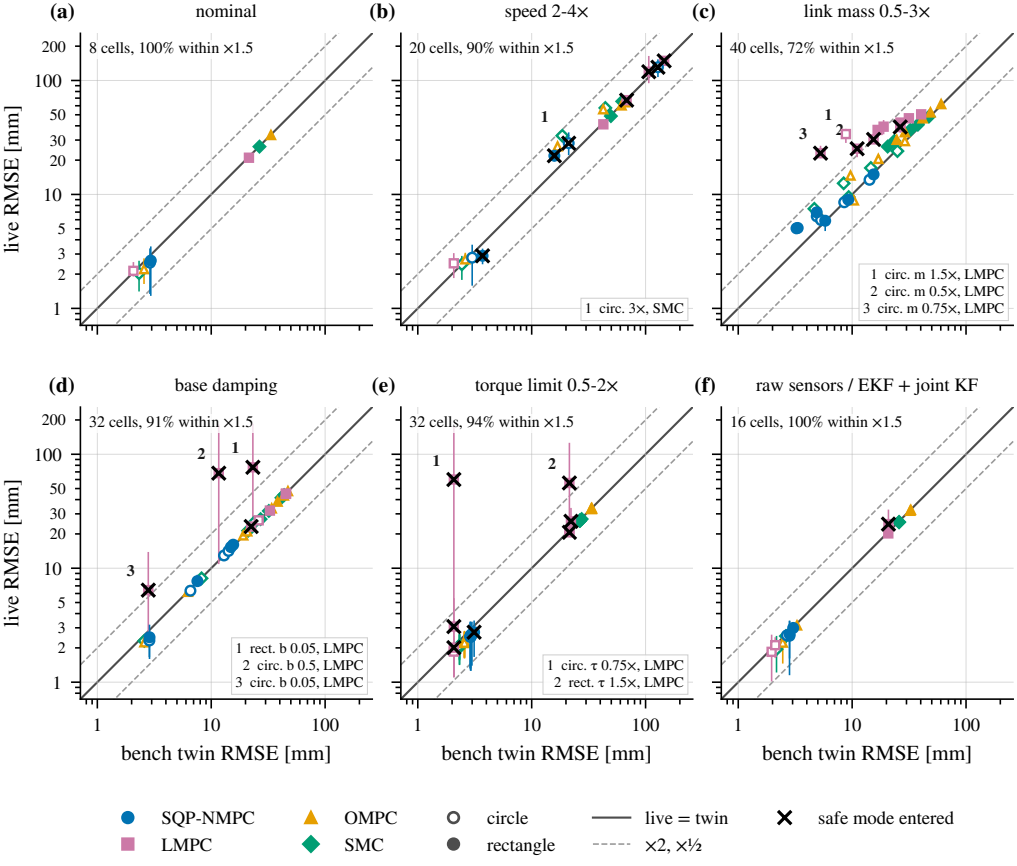
(a)
nominal
8 cells, 100% within ×1.5
(b)
speed 2-4×
20 cells, 90% within ×1.5
1 circ. 3×, SMC
(c)
link mass 0.5-3×
40 cells, 72% within ×1.5
1 circ. m 1.5×, LMPC
2 circ. m 0.5×, LMPC
3 circ. m 0.75×, LMPC
(d)
base damping
32 cells, 91% within ×1.5
1 rect. b 0.05, LMPC
2 circ. b 0.5, LMPC
3 circ. b 0.05, LMPC
(e)
torque limit 0.5-2×
32 cells, 94% within ×1.5
1 circ. τ 0.75×, LMPC
2 rect. τ 1.5×, LMPC
(f)
raw sensors / EKF + joint KF
16 cells, 100% within ×1.5
live RMSE [mm]
bench twin RMSE [mm]
SQP-NMPC
LMPC
OMPC
SMC
circle
rectangle
live = twin
×2, ×½
safe mode entered